

Lab Assignment – 2

Problem 1

Continuation of Lab 1 problem

Extend the C program developed in Lab Assignment 1 to simulate multiple independent traffic sources transmitting packets to a common destination through the same router with a bounded FIFO queue.

Each source generates packets independently according to a Poisson process. All packets share the same router queue and the same output link. The router continues to use the FIFO queue implemented in Lab Assignment 1.

Input Parameters

In addition to the parameters used in Lab Assignment 1, your program should accept:

1. Number of sources (N)
2. Packet generation rate of each source ($\lambda_1, \lambda_2, \dots, \lambda_n$). Different sources may have different packet generation rates.

Traffic Intensity

The aggregate packet arrival rate is

$$\lambda = \lambda_1 + \lambda_2 + \dots + \lambda_n$$

The traffic intensity of the output link is

$$\rho = \frac{\lambda \times L}{R}$$

where L is the packet length (bits) and R is the router-to-destination bandwidth (bits/s).

Problems

Write codes to implement the following:

1. Use four sources with equal packet generation rates. Vary the common rate so that the aggregate traffic intensity takes the values:
 $\rho = 0.2, 0.4, 0.6, 0.8, 0.9, 1.0, 1.2$
2. Use four sources with the following packet generation rates:
 1. (100, 100, 100, 700)
 2. (150, 150, 250, 450)
 3. (250, 250, 250, 250)

4. (400, 300, 200, 100)

For all cases, keep the aggregate packet generation rate approximately the same.

Output

Generate a CSV file containing:

1. Traffic intensity (ρ)
2. Aggregate packet generation rate
3. Number of packets generated
4. Number of packets delivered
5. Number of packets dropped
6. Packet-drop probability
7. Average queueing delay
8. Average end-to-end delay
9. Maximum queue occupancy

Graphs

Plot the following graphs:

1. Traffic Intensity (ρ) vs. Average Queueing Delay
2. Traffic Intensity (ρ) vs. Average End-to-End Delay
3. Traffic Intensity (ρ) vs. Packet-Drop Probability

Problem 2

Introduction to Wireshark and Network Packet Analysis

The objective of this problem is to familiarize yourself with Wireshark and understand how network communication takes place by observing real packets exchanged over the Internet. This laboratory focuses on the Network and Application Layers.

Instructions

1. Start Wireshark and select the active network interface (Ethernet in your case).
2. Start packet capture before each experiment.
3. Stop the capture after completing the experiment.
4. Save the capture as *Lab2_RollNumber.pcapng*

1. Exploring the Wireshark Interface

Start packet capture for about one minute without performing any specific activity.

Find answers to the following:

1. How many packets were captured?

2. List the different protocols observed in the Protocol column.
3. Which protocol appears most frequently?
4. Explain the purpose of the three panes in the Wireshark window.

2. ICMP Packet Analysis

Read a bit about ICMP packets.

Open a command prompt or terminal and execute:

```
ping google.com
```

Stop packet capture after five replies have been received.

Using the display filter:

```
icmp
```

Identify one ICMP Echo Request packet and its corresponding Echo Reply.

Find answers to the following:

1. What are the source and destination IP addresses?
2. What is the size of the ICMP packet?
3. What is the TTL (Time To Live) value?
4. What protocol encapsulates the ICMP message?
5. What is the time difference between the Echo Request and Echo Reply?
6. Repeat the experiment for three different websites and compare their round-trip times. Explain why they are different.

3. DNS Packet Analysis

Read a bit about DNS packets and the nslookup command

Start a new packet capture.

Open a web browser and visit a website that has not been accessed recently (or use the command `nslookup www.wikipedia.org`).

Apply the display filter:

```
dns
```

Locate the DNS Query and DNS Response packets.

Find answers to the following:

1. What domain name is being queried?
2. How many IP addresses are returned?
3. Record the returned IP address(es).

4. Which device responded to the DNS query?
5. Measure the time between the DNS query and its response.
6. Why is DNS required before a webpage can be accessed?

4. HTTP Packet Analysis

Start a new packet capture.

Visit the website: <http://neverssl.com>

Apply the display filter:

http

Locate an HTTP GET request and its corresponding response.

Find answers to the following:

1. What resource is requested by the browser?
2. What is the Host name specified in the request?
3. What HTTP version is used?
4. What is the response status code?
5. What Content-Type is returned by the server?
6. Can you view the contents of the HTTP request? Explain.

5. HTTPS Observation

Start a new packet capture.

Visit: <https://www.google.com>

Observe the captured packets.

Find answers to the following:

1. Which protocol carries the webpage data?
2. Can you view the webpage contents as you could for HTTP?
3. Which information remains visible even though the webpage is encrypted?
4. Why is HTTPS more secure than HTTP?

6. Display Filters

Apply each of the following display filters separately and record the number of packets displayed.

icmp

dns

http

ip

Find answers to the following:

1. Which filter displayed the largest number of packets?
2. Why are display filters useful when analyzing network traffic?